

Exact directional signalling for Ising, partial-SWAP, and XY unitary families

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Abstract

For a bipartite unitary U on $A \otimes B$, directional signalling from B to A is the diamond-norm distance

$$\delta_A(U) = \inf_{\mathcal{E} \in \text{CPTP}} \|\text{Tr}_B \circ \text{Ad}_U - \mathcal{E} \circ \text{Tr}_B\|_{\diamond}.$$

This quantity was introduced by Barsse, Perinotti, Tosini, and Vaglini. Exact values were subsequently obtained for the CNOT and SWAP gates, but continuous interacting families have not been evaluated. We give three such evaluations. First, for the two-qubit Ising family $U_{\theta} = \exp(-i\theta Z \otimes Z)$, we prove $\delta_A(U_{\theta}) = |\sin 2\theta|$ and identify an optimal dephasing channel. Second, for the partial-SWAP family on $\mathbb{C}^d \otimes \mathbb{C}^d$, $U_{\phi} = \cos \phi I - i \sin \phi \text{SWAP}$, unitary covariance reduces the optimization to one depolarizing parameter λ . We evaluate the fixed- λ diamond norm in closed form and thereby reduce the full problem to a scalar convex minimization. In the weak-interaction regime $0 \leq \sin \phi \leq (d^2 - 3)/(d^2 - 1)$, the result simplifies to $\delta_A(U_{\phi}) = 2 \sin \phi$; beyond this dimension-dependent threshold the optimal effective channel moves away from the identity, reaching the known SWAP value $2(1 - 1/d^2)$. Third, for the XY line $e^{-i\theta(XX+YY)}$, axial covariance gives an exact two-parameter min-max formula. Writing $S = \sin 2\theta$, the weak branch is $\sin 4\theta$, while near iSWAP the value is $S + S^2/2$, reaching $3/2$ at iSWAP. In the intervening branch the value is the unique physical root of an explicit quartic. The proofs include matching operational witnesses. We also distinguish these diamond-norm results from Hilbert–Schmidt operator-entanglement measures and provide exact-arithmetic computational certificates.

Keywords: directional signalling; diamond norm; partial SWAP; iSWAP; Ising interaction; covariant quantum channels.

1 Introduction

An interacting microscopic evolution need not induce autonomous dynamics on a subsystem. Given a unitary channel Ad_U on $A \otimes B$, one may ask how well the reduced output on A can be simulated from the reduced input on A alone. The operational answer is

$$\delta_A(U) = \inf_{\mathcal{E} \in \text{CPTP}(A)} \|\text{Tr}_B \circ \text{Ad}_U - \mathcal{E} \circ \text{Tr}_B\|_{\diamond}. \quad (1)$$

The optimization includes arbitrary AB correlations and arbitrary reference systems through the diamond norm.

Equation (1) is not a new measure. After exchanging subsystem labels, it is exactly the directional-signalling quantity introduced in Ref. [1]. That work bounded the CNOT value; the follow-up Ref. [2] computed exact CNOT and SWAP values, including parallel and asymptotic

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uses. Earlier work characterized causal and semicausal operations structurally [3, 4], while continuity of Stinespring dilations supplies general non-sharp bounds [5].

The present contribution is deliberately narrower than a new causal measure. We compute Eq. (1) sharply for three continuous unitary families. The Ising result extends the CNOT endpoint. The partial-SWAP result covers an interaction used in homogenization and collision models [9, 10] and reveals the dimension-dependent transition $\sin \phi = (d^2 - 3)/(d^2 - 1)$. The XY/iSWAP line supplies a less symmetric control: it exhibits three analytic regimes, with an algebraic quartic in the middle. A dedicated search found no previous evaluation of Eq. (1) for any of these continuous families.

The distinction from operator entanglement is important. Hilbert–Schmidt distances and the linear entropy of a vectorized unitary are useful algebraic diagnostics [8], but they do not directly give operational channel distinguishability. Every result below is stated in diamond norm and is accompanied by both an upper bound and a matching witness.

Operational picture

Informally, A is a visible system and B is hidden. They interact, B is discarded, and one tries to replace the joint evolution by the best channel acting on A alone. The number $\delta_A(U)$ is the unavoidable worst-case error of that replacement, including inputs correlated with B and with an external reference. A zero value means that A admits autonomous reduced dynamics. A nonzero value quantifies how strongly information hidden in B can affect the visible output.

Figure 1 shows the most intricate example below. The formulas change because both the optimal effective channel and the input witnessing its error change with interaction strength.

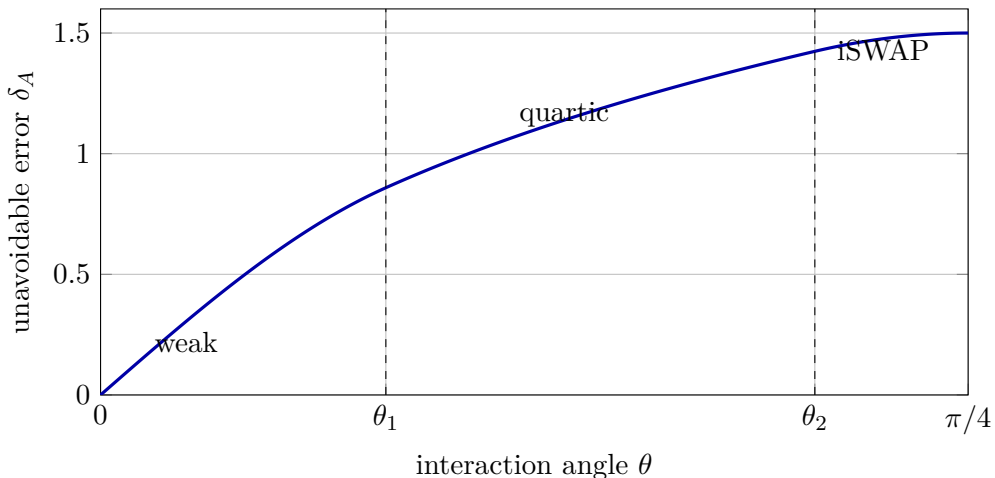


Figure 1: The exact XY directional-signalling curve. The middle segment is the unique physical root of Eq. (26).

2 Preliminaries

Let $\mathcal{L}(H)$ denote the operators on a finite-dimensional Hilbert space H . For a unitary U , write $\text{Ad}_U(X) = UXU^\dagger$. We use the unnormalized Choi operator

$$J(\Phi) = \sum_{j,k} \Phi(|j\rangle\langle k|) \otimes |j\rangle\langle k|.$$

For Hermiticity-preserving $\Phi : \mathcal{L}(X) \rightarrow \mathcal{L}(Y)$, we use the Watrous semidefinite-program characterization [6, 7]. In the unnormalized-Choi convention used here, its Hermiticity-preserving

specialization is

$$\|\Phi\|_\diamond = \max_{\rho \in \mathcal{D}(X)} \|(I_Y \otimes \sqrt{\rho})J(\Phi)(I_Y \otimes \sqrt{\rho})\|_1. \quad (2)$$

The maximizing ρ specifies a physical witness through any purification. When Φ is covariant, averaging the complete primal feasible tuple over the symmetry group shows that an invariant maximizing density may be chosen.

We use the maps

$$\mathcal{T} = \text{Tr}_B, \quad \mathcal{N}_U = \text{Tr}_B \circ \text{Ad}_U,$$

so that $\delta_A(U) = \inf_{\mathcal{E}} \|\mathcal{N}_U - \mathcal{E}\mathcal{T}\|_\diamond$. Pre- and post-composition by unitary channels are diamond-norm isometries.

Lemma 1 (Covariant twirling). *Suppose a compact group G has unitary representations H_g on $A \otimes B$ and G_g on A such that*

$$\mathcal{N}_U \text{Ad}_{H_g} = \text{Ad}_{G_g} \mathcal{N}_U, \quad \mathcal{T} \text{Ad}_{H_g} = \text{Ad}_{G_g} \mathcal{T}.$$

Then some optimal effective channel in Eq. (1) is covariant under G_g .

Proof. For a candidate $\mathcal{E}$, define $\mathcal{E}_g = \text{Ad}_{G_g^\dagger} \mathcal{E} \text{Ad}_{G_g}$. Covariance gives

$$\mathcal{N}_U - \mathcal{E}_g \mathcal{T} = \text{Ad}_{G_g^\dagger} (\mathcal{N}_U - \mathcal{E} \mathcal{T}) \text{Ad}_{H_g},$$

so every $\mathcal{E}_g$ has the same defect as $\mathcal{E}$. The Haar average $\bar{\mathcal{E}} = \int_G \mathcal{E}_g dg$ is CPTP and covariant. Convexity of the diamond norm implies that its defect is no larger. $\square$

3 The Ising family

Let

$$U_\theta = e^{-i\theta Z \otimes Z} = cI - is Z \otimes Z, \quad c = \cos \theta, \quad s = \sin \theta.$$

Theorem 1 (Exact Ising signalling). *For two qubits,*

$$\delta_A(U_\theta) = |\sin 2\theta| = 2|cs|. \quad (3)$$

An optimal effective channel is

$$\mathcal{E}_\theta(X) = c^2 X + s^2 Z X Z.$$

Proof. Define

$$M(X) = \text{Tr}_B[(I \otimes Z)X], \quad L(X) = -\frac{i}{2}[Z, M(X)].$$

Expanding $U_\theta X U_\theta^\dagger$ and using cyclicity inside the partial trace gives

$$\mathcal{N}_{U_\theta} - \mathcal{E}_\theta \mathcal{T} = 2cs L. \quad (4)$$

The completely bounded trace norm of M is one: its adjoint is the complete operator-norm isometry $Y \mapsto Y \otimes Z$. Also $K \mapsto -i[Z, K]/2$ has completely bounded trace norm at most one by the triangle inequality for left and right unitary multiplication. Hence $\|L\|_\diamond \leq 1$, which proves the upper bound $2|cs|$.

For the lower bound, consider

$$\rho_0 = |+\rangle\langle+|_A \otimes |0\rangle\langle 0|_B, \quad \rho_1 = |+\rangle\langle+|_A \otimes |1\rangle\langle 1|_B.$$

They have the same A marginal, so every candidate $\mathcal{E}$ assigns them the same simulated output. Their true reduced outputs differ by $2cs Y$, whose trace norm is $4|cs|$. The triangle inequality therefore forces at least one of the two simulation errors to be at least $2|cs|$. This matches the upper bound. $\square$

At $\theta = \pi/4$, U_θ is locally equivalent to CZ and CNOT. Equation (3) then reproduces the exact CNOT value one, already obtained in Ref. [2]. The continuous formula, rather than that endpoint, is the result of Theorem 1.

4 The partial-SWAP family in dimension d

Let $A \cong B \cong \mathbb{C}^d$ and consider

$$U_\phi = cI - isS, \quad S = \text{SWAP}, \quad c = \cos \phi, \quad s = \sin \phi, \quad (5)$$

for $0 \leq \phi \leq \pi/2$. For qubits, the equal-angle Cartan interaction is this family up to global phase:

$$e^{-it(XX+YY+ZZ)} = e^{it}e^{-2itS}, \quad \phi = 2t.$$

4.1 Reduction to depolarizing channels

The unitary U_ϕ commutes with $V \otimes V$ for every $V \in U(d)$. Lemma 1 therefore permits an optimal effective channel to be chosen covariant under all unitary conjugations. Every such channel is depolarizing:

$$\mathcal{E}_\lambda(X) = \lambda X + (1 - \lambda) \text{Tr}(X) \frac{I}{d}, \quad -\frac{1}{d^2 - 1} \leq \lambda \leq 1. \quad (6)$$

This interval is precisely the CPTP range. Direct expansion gives

$$\mathcal{N}_{U_\phi}(X) = c^2 \text{Tr}_B X + s^2 \text{Tr}_A X - ics \text{Tr}_B(SX - XS),$$

where $\text{Tr}_A X$ is relabelled onto the output copy of A .

4.2 The fixed-channel diamond norm

Define

$$\begin{aligned} A &= 1 - \lambda, & B &= A^2 + d^2 \lambda s^2, \\ H_0 &= \frac{d(d^2 - 3)}{d^2 - 1}, & H_1 &= \frac{2}{d^2 - 1}. \end{aligned} \quad (7)$$

For $B > 0$, also put

$$Q = A^2(H_0^2 - H_1^2) + 4B, \quad C = 1 - \frac{A^2 H_1^2}{4B}, \quad D = \frac{AH_0 + \sqrt{Q}}{C}. \quad (8)$$

Lemma 2 (Fixed depolarizing channel in dimension d). *Let*

$$d_{d,\phi}(\lambda) = \left\| \mathcal{N}_{U_\phi} - \mathcal{E}_\lambda \mathcal{T} \right\|_\diamond.$$

Apart from the identity point $B = 0$, where the norm vanishes,

$$d_{d,\phi}(\lambda) = \begin{cases} D/d, & AH_1 D \leq 4B, \\ 2A(H_0 + H_1)/d, & AH_1 D > 4B. \end{cases} \quad (9)$$

The quantities B and C are positive and Q is nonnegative on the CPTP interval away from that identity point, so the closed form is well defined.

Proof. The Watrous SDP is invariant under the simultaneous $V \otimes V$ covariance action. Averaging its complete primal feasible tuple therefore preserves both feasibility and objective value, and Schur–Weyl duality permits an optimal input density

$$\rho = xP_- + yP_+, \quad r_-x + r_+y = 1, \quad r_\pm = \frac{d(d \pm 1)}{2}.$$

Let $u = r_+ y$ be its total symmetric-subspace weight. Put

$$P = \frac{(d-1)(d+2)}{d+1}, \quad R = \frac{(d+1)(d-2)}{d-1}, \quad \tilde{B} = (1+\lambda)^2 - 4\lambda c^2.$$

For

$$K_\rho = (I \otimes \sqrt{\rho}) J(\mathcal{N}_{U_\phi} - \mathcal{E}_\lambda \mathcal{T})(I \otimes \sqrt{\rho}),$$

the Schur–Weyl blocks give the spectrum

eigenvalue	multiplicity
$-(1-\lambda)y/d$	$d(d-1)(d+2)/2$
$-(1-\lambda)x/d$	$d(d+1)(d-2)/2$
ν_+, ν_-	d each

where

$$\begin{aligned} \nu_+ + \nu_- &= \frac{1-\lambda}{d^2} [Pu + R(1-u)], \\ \nu_+ \nu_- &= \frac{u(1-u)}{d^2} \left[\frac{(1-\lambda)^2}{d^2} PR - \tilde{B} \right]. \end{aligned} \quad (10)$$

The last product is nonpositive. Summing all eigenvalue magnitudes gives

$$d_{d,\phi}(\lambda) = \max_{0 \leq u \leq 1} \left\{ \frac{1-\lambda}{d} [Pu + R(1-u)] + \sqrt{\frac{(1-\lambda)^2}{d^2} [Pu - R(1-u)]^2 + 4u(1-u)\tilde{B}} \right\}. \quad (11)$$

With $t = 2u - 1$, elementary algebra turns Eq. (11) into

$$\max_{-1 \leq t \leq 1} \frac{1}{d} \left[A(H_0 + H_1 t) + \sqrt{A^2(H_0 + H_1 t)^2 + 4B(1-t^2)} \right]. \quad (12)$$

If the maximum is interior, its location is $t_* = AH_1 D / (4B)$ and its value is D/d . The condition $t_* \leq 1$ gives the first branch of Eq. (9); otherwise the maximum is at $t = 1$ and gives its second branch. The maximizing ρ , together with any purification, realizes the matching lower-bound witness; the symmetrized SDP calculation supplies the upper bound. $\square$

4.3 Optimization and the dimension-dependent threshold

Theorem 2 (Partial-SWAP signalling in arbitrary dimension). *For every $d \geq 2$,*

$$\delta_A(U_\phi) = \min_{-1/(d^2-1) \leq \lambda \leq 1} d_{d,\phi}(\lambda), \quad (13)$$

with $d_{d,\phi}$ given explicitly by Eq. (9). Moreover,

$$0 \leq \sin \phi \leq \frac{d^2 - 3}{d^2 - 1} \implies \lambda_* = 1, \quad \delta_A(U_\phi) = 2 \sin \phi. \quad (14)$$

For stronger non-SWAP coupling, a minimizer lies in $0 < \lambda_* < 1$. At $\phi = \pi/2$,

$$\lambda_* = 0, \quad \delta_A(S) = 2 \left(1 - \frac{1}{d^2} \right). \quad (15)$$

Table 1: Dimension dependence of the partial-SWAP weak threshold and the known full-SWAP endpoint.

d	weak threshold for $\sin \phi$	$\delta_A(S)$
2	1/3	3/2
3	3/4	16/9
4	13/15	15/8
d	$(d^2 - 3)/(d^2 - 1)$	$2(1 - 1/d^2)$

Proof. Equation (13) follows from covariance and Lemma 2. The function $d_{d,\phi}(\lambda)$ is convex because it is the diamond norm of a map affine in λ . At $\lambda = 1$,

$$d_{d,\phi}(1) = 2s, \quad \left. \frac{d}{d\lambda} d_{d,\phi}(\lambda) \right|_{1^-} = s - \frac{d^2 - 3}{d^2 - 1}.$$

Convexity proves Eq. (14). The matching invariant density has equal total symmetric and antisymmetric weights,

$$\rho = \frac{dI - S}{d(d^2 - 1)}.$$

At $\lambda = 0$ and $c > 0$, the derivative is $-2(d^2 - 1)c^2/d^2 < 0$. Convexity excludes negative λ from the outer optimum, while the positive derivative at $\lambda = 1$ in the strong regime places a minimizer in $(0, 1)$. At SWAP the minimizer is $\lambda = 0$; Eq. (12) is maximized at $t = 1/d$, corresponding to $\rho = I/d^2$, and yields Eq. (15), in agreement with Ref. [2]. $\square$

5 The XY/iSWAP line

Consider

$$U_\theta = e^{-i\theta(XX+YY)}, \quad 0 \leq \theta \leq \frac{\pi}{4}, \quad C = \cos 2\theta, \quad S = \sin 2\theta. \quad (16)$$

The zero- and two-excitation states are fixed, while on $\text{span}\{|01\rangle, |10\rangle\}$ the unitary is $CI - iSX$. Equivalently,

$$U_\theta = \cos^2 \theta II - \frac{i}{2} \sin 2\theta (XX + YY) + \sin^2 \theta ZZ.$$

5.1 Axial covariance and the fixed-channel norm

Joint Pauli covariance first restricts an optimal effective channel to a Pauli channel. The additional axial symmetry generated by $ZI + IZ$ makes its X and Y weights equal. We therefore write

$$\mathcal{E}_{u,w} : (r_x, r_y, r_z) \mapsto (ur_x, ur_y, wr_z), \quad -1 \leq w \leq 1, \quad |u| \leq \frac{1+w}{2}. \quad (17)$$

The corresponding Pauli probabilities are

$$q_I = \frac{1+2u+w}{4}, \quad q_X = q_Y = \frac{1-w}{4}, \quad q_Z = \frac{1+w-2u}{4}.$$

Lemma 3 (XY fixed-channel norm). *For fixed (u, w) in Eq. (17), put $B = \frac{1}{2} - A$ and*

$$\alpha = \frac{A(1-w)}{2}, \quad \gamma^2 = \frac{AB}{2}, \quad g = C - u, \quad h = S, \quad \beta = \frac{B}{2} [w - (C^2 - S^2) + 2iSC].$$

Define

$$M_3(A) = \begin{pmatrix} \alpha & \gamma(g+ih) & \gamma(-g+ih) \\ \gamma(g-ih) & 0 & \beta \\ \gamma(-g-ih) & \bar{\beta} & 0 \end{pmatrix}. \quad (18)$$

Then

$$\|\mathcal{N}_{U_\theta} - \mathcal{E}_{u,w}\mathcal{T}\|_\diamond = \max_{0 \leq A \leq 1/2} 2[\alpha + \|M_3(A)\|_1]. \quad (19)$$

Proof. Average the complete primal diamond-SDP tuple over joint axial rotations and $X \otimes X$. The residual odd-sector coherence changes sign under the real-linear involution

$$X \mapsto (Z \otimes I)X^*(Z \otimes I),$$

which is a covariance because $U_\theta(Z \otimes I) = (Z \otimes I)U_\theta^*$. Averaging this involution leaves an optimal density of the form

$$\rho = A(|00\rangle\langle 00| + |11\rangle\langle 11|) + B(|01\rangle\langle 01| + |10\rangle\langle 10|).$$

Inserting ρ in Eq. (2), the scaled Choi operator splits into two unitarily equivalent blocks. Each has one scalar eigenvalue $-\alpha$ and the three eigenvalues of Eq. (18), which proves Eq. (19). For reference, the characteristic polynomial $x^3 + p_2x^2 + p_1x + p_0$ of M_3 has

$$\begin{aligned} p_2 &= -\alpha, \\ p_1 &= -\left[2\gamma^2(g^2 + h^2) + |\beta|^2\right], \\ p_0 &= \alpha|\beta|^2 + 2\gamma^2(g^2 - h^2)\Re\beta - 4\gamma^2gh\Im\beta. \end{aligned}$$

□

Lemma 3 is an exact finite-dimensional answer along the whole XY line:

$$\delta_A(U_\theta) = \min_{\substack{-1 \leq w \leq 1 \\ |u| \leq (1+w)/2}} \max_{0 \leq A \leq 1/2} 2[\alpha + \|M_3(A)\|_1]. \quad (20)$$

Whenever M_3 has one nonnegative and two nonpositive eigenvalues, $\text{Tr } M_3 = \alpha$ gives $2[\alpha + \|M_3\|_1] = 4\lambda_{\max}(M_3)$. Two coupling regimes simplify further.

5.2 Closed weak and iSWAP branches

Let $\theta_1 = 0.258270520262\dots$ be the unique root in $(0, \pi/4)$ of

$$2SC = 3C^3 + C^2 - C - 1, \quad (21)$$

and let $\theta_2 = 0.646615513406\dots$ solve

$$1 + C^2 = 2C(1 + S). \quad (22)$$

Equivalently, $\tan \theta_2$ is the positive root of $t^3 + t^2 - 1 = 0$.

Theorem 3 (Closed XY signalling branches). *For $0 \leq \theta \leq \theta_1$,*

$$\delta_A(U_\theta) = \sin 4\theta, \quad (q_I, q_X, q_Y, q_Z) = \left(C^2, \frac{S^2}{2}, \frac{S^2}{2}, 0\right). \quad (23)$$

For $\theta_2 \leq \theta \leq \pi/4$,

$$\delta_A(U_\theta) = S + \frac{S^2}{2}, \quad (24)$$

with

$$(q_I, q_X, q_Y, q_Z) = \frac{1}{4} \left(1 + C^2 + 2C(1 + S), S^2, S^2, 1 + C^2 - 2C(1 + S)\right). \quad (25)$$

In particular, $\delta_A(\text{iSWAP}) = 3/2$.

Proof. For every effective channel, the choice $A = 0$ in Eq. (19) gives

$$\sqrt{[w - (C^2 - S^2)]^2 + 4S^2C^2} \geq 2SC.$$

This is the weak lower bound. For the channel in Eq. (23), let $H_1 = (SC/2)I - M_3(A)$. Its leading principal minor is $S(C - 2AS)/2$, and its $(2, 3)$ principal minor is $AC^2(A - 1)(C - 1)(C + 1)$. The remaining order-two minor is nonnegative after completing its quadratic square. Finally,

$$\det H_1 = \frac{AC(C - 1)}{4}K(A),$$

where K is convex and

$$K(0) = S[2SC - (3C^3 + C^2 - C - 1)], \quad K(1/2) = -C(C + 1)S(C - S).$$

Both endpoint values are nonpositive precisely through the weak interval; convexity gives $K(A) \leq 0$. Moreover,

$$\det M_3 = -\frac{AC^2(2A - 1)^2(C - 1)^3(C + 1)}{4} \geq 0,$$

while the sum of its order-two principal minors is nonpositive. Thus M_3 has one nonnegative and two nonpositive eigenvalues. The principal minors above give $H_1 \geq 0$, hence $\lambda_{\max}(M_3) \leq SC/2$; together with the inertia identity following Eq. (20), this makes Eq. (19) at most $2SC$. This proves Eq. (23) and identifies the first threshold.

For the strong branch, Eq. (25) has $u = C(1 + S)$ and $w = C^2$. It is CPTP exactly when Eq. (22) holds with the strong-side inequality. At $A = 1/4$, the largest eigenvalue of M_3 is

$$L = \frac{S(2 + S)}{8},$$

with eigenvector proportional to $(i\sqrt{2}, 1, 1)$. The other two eigenvalues have sum $-S/4$ and product $S^2(7S^2 + 2S - 8)/64$, so they are negative throughout this regime. The derivatives of L in both u and w vanish. Convexity of the trace norm in the channel parameters therefore proves that the fixed $A = 1/4$ witness has global minimum $4L = S + S^2/2$.

It remains to prove the matching upper bound. Put $H_3 = LI - M_3(A)$. Direct factorization gives

$$\det H_3 = -\frac{S^3(4A - 1)^2(S + 2)}{512} \left[A(28S^2 + 8S - 32) + 12 - 13S^2 - 4S \right].$$

The bracket is negative at both endpoints of $A \in [0, 1/2]$. The two nontrivial order-two principal minors are $S^2G_1/64$ and $S^2G_2/64$, where, with $t = 2A$,

$$\begin{aligned} G_1 &= (16 - 8S^2)t^2 + (6S^2 - 4S - 16)t + (S + 2)^2, \\ G_2 &= (12S^2 - 16)t^2 + (32 - 24S^2)t + 13S^2 + 4S - 12. \end{aligned}$$

G_1 has positive completed-square minimum, while G_2 is concave with nonnegative endpoint values $13S^2 + 4S - 12$ and $(S + 2)^2$. Together with $(H_3)_{11} = S(S + 2 - 4AS)/8$, all principal minors are nonnegative, so $H_3 \geq 0$. In addition,

$$\det M_3 = \frac{AS^3(2A - 1)^2(S + 2)(7S^2 + 2S - 8)}{32} \geq 0,$$

and the sum of its order-two principal minors is nonpositive. Thus again M_3 has one nonnegative and two nonpositive eigenvalues, and Eq. (19) is at most $4L$. This proves Eq. (24). At $S = 1$, the four probabilities are $1/4$ and the value is $3/2$. $\square$

Theorem 4 (Intermediate XY branch). *For $\theta_1 < \theta < \theta_2$, put $t = \tan \theta$. Then $\delta_A(U_\theta)$ is the unique root $\Delta \in (1/2, 2)$ of*

$$\begin{aligned} Q_+(\Delta, t) = & (t^6 + t^5 + 3t^4 + 2t^3 + 3t^2 + t + 1)\Delta^4 \\ & + (2t^7 - 46t^6 - 122t^5 - 170t^4 - 138t^3 - 58t^2 - 14t + 2)\Delta^3 \\ & + (-84t^7 + 24t^6 + 72t^5 + 96t^4 - 68t^3 - 56t^2 - 32t)\Delta^2 \\ & + (96t^7 + 64t^6 + 160t^5 + 384t^4 + 320t^3 + 128t^2)\Delta \\ & + 64t^5(2t^2 + 2t + 1). \end{aligned} \quad (26)$$

Proof. We first prove the active facet. On $q_Z = 0$, write $w = 2u - 1$, $B = 1/2 - A$, and let λ be the largest eigenvalue of $M_3(A)$. Put

$$m = (u - C)^2 + S^2, \quad N = (u - C^2)^2 + C^2 S^2, \quad \Theta = u(1 - u)(1 - C)^2.$$

The characteristic polynomial becomes

$$P(\lambda; A, u) = \lambda^3 - A(1 - u)\lambda^2 - (ABm + B^2N)\lambda - AB^2\Theta. \quad (27)$$

Thus $\det M_3 = AB^2\Theta \geq 0$, while the sum of its order-two principal minors is $-(ABm + B^2N) \leq 0$. In the interior M_3 has one positive and two negative eigenvalues, so the fixed-witness objective is 4λ and $P_\lambda > 0$.

For fixed channel parameters, the diamond-SDP value is concave in its input density: the feasible pairs (ρ, Y) in the Watrous primal form a convex set and the objective is linear. Since the invariant density is affine in A , the objective is concave in A . Its maximizer is unique, since otherwise an interval of maximizing A values would make the cubic $P(\lambda; A, u)$ vanish identically despite its leading coefficient $-\Theta$. The resulting facet objective $f(u)$ is convex and continuous. Its endpoint derivatives point inward, and its minimizer has $0 < u < 1$ and $0 < A < 1/2$ in the intermediate interval. At $A = 0$ the only possible stationary value is $u = C^2$, where the inward derivative has sign

$$C(1 - C)S[2SC - (3C^3 + C^2 - C - 1)],$$

which changes precisely at θ_1 .

A flat interval of minimizing u values is impossible. Indeed,

$$\text{disc}_A P = \frac{(C - 1)^2 \lambda^2 (2\lambda + u)}{16} R(u, \lambda, C), \quad [u^7]R = (1 - C)^2 - 8\lambda.$$

If $A = 0$ on a subinterval, the boundary value $2\sqrt{(u - C^2)^2 + C^2 S^2}$ is strictly convex and cannot be constant. Hence any flat interval contains an open subinterval with $0 < A < 1/2$, where the displayed discriminant applies. Constancy on an interval would force R to vanish identically in u , hence $4\lambda = (1 - C)^2/2 < 1/2$, whereas the $A = 0$ witness gives $4\lambda \geq 2SC > 1/2$. The facet saddle is therefore unique and varies continuously.

It remains to check the normal KKT multiplier. Holding w fixed before restricting to the facet gives

$$P_u = 2ABG, \quad G = (C - u)\lambda + B\Psi, \quad \Psi = u^2 - C(1 + C)u + C.$$

Here $\Psi > 0$ because its discriminant is $C(C - 1)(C^2 + 3C + 4) < 0$. Eliminating A, u, λ from $P = P_A = P_u + 2P_w = G = 0$, after saturating the interior factors, gives

$$5C^3 + C^2 + 3C - 1 = 0.$$

This is exactly the second-threshold equation. At θ_1 the limiting weak saddle has $A = 0$, $u = C^2$, $\lambda = SC/2$, and

$$G = \frac{C(1 - C)}{2}[(1 + C) + SC] > 0.$$

Therefore $G > 0$ throughout (θ_1, θ_2) . The gradient of the actual convex channel objective is a positive multiple of the inward normal to $q_Z \geq 0$, so the convex KKT conditions prove that this facet saddle is the global optimum over the CPTP triangle.

We next identify its algebraic sheet. Put $a = 2A$ and $F = 64(1+t^2)^2 P(\Delta/4; a/2, u)$. Write $F = P_2 u^2 + P_1 u + P_0$. At the saddle $F_u = 0$, and direct factorization gives

$$\text{disc}_u F = 4W_+(a, \Delta, t)W_+(a, \Delta, -t),$$

where

$$W_+ = (1+t^2)^2 \left[a \left(\Delta + \frac{4(1-a)t^2}{1+t^2} \right)^2 - 2\Delta^2 + \frac{8\Delta(1-a)t}{1+t^2} \left(\frac{2at^2+1-t^2}{1+t^2} \right) \right].$$

For $t \in [1/4, 19/25]$, $\Delta \geq 1/2$, and $0 \leq a \leq 1$, an elementary rearrangement gives

$$-\frac{W_+(a, \Delta, -t)}{(1+t^2)^2} \geq \frac{171381}{972196} > 0.$$

Thus the physical saddle lies on $W_+ = 0$, without numerical sheet selection.

The second stationary equation, after using the linear equation $F_u = 0$, is $G_8(a, \Delta, t) = 0$. If P_k denote the coefficients of F above, it can be defined without expansion by

$$-16G_8 = (\partial_a P_2)P_1^2 - 2P_1(\partial_a P_1)P_2 + 4(\partial_a P_0)P_2^2.$$

Exact elimination of a gives

$$\begin{aligned} \text{Res}_a(W_+, G_8) &= -2^{36}\Delta^{12}t^{28}(1+t^2)^{10}E_1E_2^2Q_+(\Delta, t), \\ E_1 &= \Delta^2(1+3t^2) + 4(1-t^2)(1+\Delta), \\ E_2 &= \Delta(1+t^2)^2 + 2(1+t)^2(1+2t-t^2). \end{aligned}$$

Both E_1 and E_2 are strictly positive for $0 < t < 1$ and $\Delta > 0$, proving $Q_+(\Delta, t) = 0$.

Finally,

$$\begin{aligned} Q_+(1/2, t) &= \frac{1}{16}(2484t^7 + 2565t^6 + 2349t^5 + 3119t^4 + 2014t^3 + 687t^2 - 155t + 5) > 0, \\ Q_+(2, t) &= -32(9t^5 + 5t^4 + 22t^3 + 12t^2 + 7t - 1) < 0 \end{aligned}$$

on $[1/4, 19/25]$, while

$$\text{disc}_\Delta Q_+ = 2^{16}t^7(1+t^2)(t^3-t^2-t-1)^2N_9(t)^3$$

with

$$N_9(t) = 27t^9 + 378t^8 + 1026t^7 + 2106t^6 + 2772t^5 + 2646t^4 + 1658t^3 + 678t^2 + 141t + 16.$$

The discriminant is nonzero there. Hence the root count in $(1/2, 2)$ is constant, and a Sturm count at $t = 1/2$ gives one. Direct substitution of the weak and strong values produces the squared threshold factors

$$(t^6 - 2t^5 - 3t^4 + 7t^2 + 2t - 1)^2, \quad (t^3 + t^2 - 1)^2,$$

respectively, with strictly positive cofactors. The middle root therefore joins both closed branches exactly. $\square$

The full curve satisfies $\delta_A(\theta) = \delta_A(\pi/2 - \theta)$ and returns to zero at $\theta = \pi/2$, where $U_{\pi/2} = Z \otimes Z$.

6 Discussion

The three families exhibit different mechanisms. In the Ising family, one dephasing channel is optimal for every coupling, and two product inputs already witness the exact defect. Along the partial-SWAP family, the optimal effective channel itself changes: it remains the identity until $\sin \phi = (d^2 - 3)/(d^2 - 1)$, then moves into the depolarizing interval and reaches complete depolarization at SWAP. The optimal diamond witness likewise generally uses a reference system, as encoded by a purification of the symmetric/antisymmetric density in Lemma 2.

The XY line has less symmetry and a genuine active-set phase diagram. Its weak branch is controlled by the one-excitation witness, its central iSWAP branch by the maximally mixed witness, and the intervening interval moves through an algebraic saddle. The iSWAP value $3/2$ equals the qubit SWAP value despite the gates belonging to different local-equivalence classes.

The threshold is operational rather than spectral: it is where the outer channel optimization first benefits from adding depolarization. It is not visible from operator entanglement alone. This supplies a concrete example in which a coarse-grained autonomy question differs from a Hilbert–Schmidt interaction diagnostic.

The result is limited in three important ways. First, Eq. (1) is a pre-existing signalling measure, not a newly defined quantity. Second, the exact CNOT and SWAP endpoints are prior results [2]; the contribution here is the continuous-family evaluation. Third, a closed formula for a generic two-qubit Cartan interaction $e^{-i(\alpha XX + \beta YY + \gamma ZZ)}$ remains open. Joint-Pauli symmetry reduces the latter problem to optimization over the Pauli-channel tetrahedron, but the three leakage sectors interfere in trace norm and the partial-SWAP $U(2)$ symmetry is absent.

7 Reproducibility

The accompanying repository implements finite-dimensional certificates over the Gaussian rationals $\mathbb{Q}(i)$. It verifies the Ising residual factorization, the physical product-state witness, the equal-Cartan/partial-SWAP identity, exact qubit, qutrit, and dimension-four weak-branch values, the negative depolarizing boundary, the SWAP endpoint, rational points on both closed XY branches including iSWAP, and the middle-branch quartic controls. The focused suite is run from the repository root with

```
python3 -m unittest discover -s tests -p 'test_aqc_*.py' -v
```

The larger symbolic identities in Theorem 4 can be regenerated independently with the optional proof dependencies:

```
python3 -m pip install -e '[proof]'
python3 tools/derive_xy_middle.py
```

The plotted data are generated without third-party Python dependencies:

```
python3 paper/generate_xy_plot.py
```

The code is a certificate and regression harness; the analytic proofs above do not depend on floating-point optimization. The exact harness covers an interior point of the weak branch and the SWAP boundary of the strong branch; the generic interior strong-branch result is certified analytically by Eqs. (10)–(13), not by encoding its generally algebraic minimizer in the rational backend.

8 Conclusion

Directional signalling can be evaluated sharply beyond isolated gates. For two-qubit Ising interactions it is $|\sin 2\theta|$. In every finite dimension, partial SWAP reduces to an explicit scalar

convex minimization with weak branch $2 \sin \phi$, transition $(d^2 - 3)/(d^2 - 1)$, and known SWAP endpoint $2(1 - 1/d^2)$. The calculation of the XY line gives an exact three-regime curve, including the new iSWAP endpoint $3/2$ and a unique quartic root in the intermediate regime. These results provide controlled settings in which an optimal autonomous effective channel changes nonperturbatively with interaction strength. The axially symmetric XXZ surface is the next tractable step toward the full Cartan chamber.

A Audit details for the intermediate XY branch

This appendix records the boundary and saturation details used in Theorem 4. They are included to make the exact eliminations auditable without reconstructing the branch selection from numerical output.

A.1 Boundary exclusion and uniqueness

On the facet $w = 2u - 1$, Eq. (27) satisfies

$$P_A|_{A=1/2, \lambda=(1-u)/2} = \frac{u(1-u)(1-C)}{2} > 0.$$

Since $P_\lambda > 0$, the largest eigenvalue decreases as A approaches $1/2$; that endpoint cannot maximize the witness objective. At $A = 0$, the objective is

$$2\sqrt{(u - C^2)^2 + C^2 S^2}.$$

Its only stationary point is $u = C^2$. The derivative into $A > 0$ there has the sign of

$$C(1 - C)S[2SC - (3C^3 + C^2 - C - 1)],$$

which is positive exactly after the first threshold. Thus the middle saddle has $0 < A < 1/2$. Direct endpoint evaluation gives $f'(0^+) = -2$ and $f'(1^-) = 2S$, so also $0 < u < 1$.

For fixed u , concavity in A makes the maximizer set an interval. Any nontrivial interval would make the cubic $P(\lambda; A, u)$ vanish identically in A , contradicting the coefficient $-\Theta$ of A^3 . For fixed C , convexity in u makes the minimizer set an interval. A flat portion on the $A = 0$ boundary is excluded by strict convexity of the displayed square root. In the interior it would force the A -discriminant polynomial R in the main proof to vanish identically in u , including its leading coefficient $(1 - C)^2 - 8\lambda$. This contradicts $4\lambda \geq 2SC > 1/2$. Hence both optimizers are unique, and Berge's theorem gives their continuity in C .

A.2 Multiplier elimination and saturation ledger

The zero-multiplier calculation starts from the unrestricted characteristic polynomial and the equations

$$P = 0, \quad P_A = 0, \quad P_u + 2P_w = 0, \quad G = 0.$$

The physical interior permits saturation by

$$A, B, u, 1 - u, C, 1 - C^2, \lambda, P_\lambda, \Psi.$$

Every factor is nonzero: the boundary factors were excluded above; $0 < C < 1$ and $\lambda > 0$ hold in the middle interval; $P_\lambda > 0$ follows from the simple positive eigenvalue; and $\Psi > 0$ follows from its negative discriminant. The substitution $B = (u - C)\lambda/\Psi$ therefore loses no physical solution. The remaining generic component $u = -2\lambda$ is incompatible with $u, \lambda > 0$. After these exclusions, the saturated elimination ideal in $\mathbb{Q}[C]$ is generated by $5C^3 + C^2 + 3C - 1$. It vanishes only at θ_2 , establishing the nonvanishing multiplier used in the KKT argument. The optional symbolic script also prints the unsaturated resultant factors and the justification for excluding each one.

A.3 Physical sheet and root isolation

With $\sigma = t^2/(1+t^2)$ and $b = 1 - a$, the conjugate sheet obeys the exact rearrangement

$$-\frac{W_+(a, \Delta, -t)}{(1+t^2)^2} = (1+b)\Delta^2 + 4b\Delta\kappa - 16(1-b)b^2\sigma^2,$$

where

$$\kappa = SC - 2a\sigma(1-S) \geq S + C - 1 > 0.$$

Using $a(1-a) \leq 1/4$, $\Delta \geq 1/2$, and $t \in [1/4, 19/25]$ gives the rational lower bound 171381/972196. Thus the conjugate sheet cannot contain the physical saddle. The resultant in the main proof then forces $Q_+ = 0$ because all its remaining factors are strictly positive.

The leading coefficient of Q_+ is

$$(1+t^2)^2(t^2+t+1) > 0.$$

The endpoint signs and nonzero discriminant displayed in the main proof prevent roots from entering, leaving, colliding, or escaping to infinity on the enclosing interval. The exact Sturm variation difference at $t = 1/2$ is one between $\Delta = 1/2$ and $\Delta = 2$. Hence the physical root is unique throughout the middle regime.

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